

Transformate 1: Fourier in $\mathcal{S} = \mathcal{S}(\mathbb{R}^n)$.

Def. $\mathcal{S}(\mathbb{R}^n) := \{u \in C^\infty(\mathbb{R}^n) \text{ tale che}$
 $\forall \alpha, \beta \text{ multi-indici sup } |x^\alpha \mathcal{D}^\beta u(x)| < +\infty\}$
 $x \in \mathbb{R}^n$

Esempio:

$$C_0^\infty(\mathbb{R}^n) \subsetneq \mathcal{S}(\mathbb{R}^n) \subsetneq C^\infty(\mathbb{R}^n)$$

$\uparrow$ $\nwarrow$

$$u(x) \in \mathcal{S}(\mathbb{R}^n) \setminus C_0^\infty(\mathbb{R}^n)$$

$$u(x) = e^{-x^2} \quad n=1$$

Osservazioni su $\mathcal{S}(\mathbb{R}^n)$

- $u \in \mathcal{S}(\mathbb{R}^n) \Rightarrow x^\gamma u \in \mathcal{S}(\mathbb{R}^n)$
- (E.g. $n=2$ $x = (x_1, x_2)$ $\alpha = (\alpha_1, \alpha_2)$ $x^\alpha = x_1^{\alpha_1} x_2^{\alpha_2}$)
- $u \in \mathcal{S}(\mathbb{R}^n) \Rightarrow \mathcal{D}^\gamma u \in \mathcal{S}(\mathbb{R}^n)$
- $\mathcal{S}(\mathbb{R}^n)$ è uno spazio vettoriale
- $\mathcal{S}(\mathbb{R}^n) \subseteq L^1(\mathbb{R}^n)$

$$\underline{n=1} \quad |x^2 u(x)| \leq M \Rightarrow |u(x)| \leq \frac{M}{x^2} \in L^1(\mathbb{R})$$

• Valgono per ogni $u \in \mathcal{S}(\mathbb{R}^n)$;

$$(1) \widehat{D^\alpha u} = i^{|\alpha|} \xi^\alpha \widehat{u}(\xi)$$

$$(2) D^\alpha \widehat{u} = (-i)^{|\alpha|} \widehat{x^\alpha u}$$

Proposizione: Se $u \in \mathcal{S}(\mathbb{R}^n)$, allora $\widehat{u} \in \mathcal{S}(\mathbb{R}^n)$

e vale
$$\boxed{\check{u} = \frac{1}{(2\pi)^n} \widehat{\widehat{u}}}$$

Dim. Basta far vedere che

$$u \in \mathcal{S}(\mathbb{R}^n) \Rightarrow \widehat{u} \in \mathcal{S}(\mathbb{R}^n).$$

(In tal caso: $u \in \mathcal{S} \Rightarrow u \in L^1$, $\widehat{u} \in \mathcal{S} \Rightarrow \widehat{u} \in L^1$)

• $\widehat{u} \in \mathcal{C}^\infty(\mathbb{R}^n)$ perché $x^\alpha u(x) \in L^1(\mathbb{R}^n) \forall \alpha$

$$(u \in \mathcal{S} \Rightarrow x^\alpha u \in \mathcal{S} \subseteq L^1)$$

applico la Prop. 2.

• $\sum \widehat{D^\beta u}(\xi)$ è limitata, perché è la trasf. di Fourier di una funzione L^1 ($\mathcal{F}: L^1 \rightarrow L^\infty$)

$$\begin{aligned} \sum \widehat{D^\beta u} &\stackrel{(2)}{\sim} \sum \widehat{x^\beta u} &\stackrel{(1)}{\sim} \widehat{D^\alpha (x^\beta u)} \\ &\uparrow &\uparrow &\uparrow L^1 \end{aligned}$$

$$u \in \mathcal{S} \Rightarrow x^\beta u \in \mathcal{S} \Rightarrow \mathcal{D}^\alpha(x^\beta u) \in \mathcal{S} \subseteq L^2 \quad \square$$

Altre proprietà di $\mathcal{F}$ in $\mathcal{S}(\mathbb{R}^n)$

$$\circ \int_{\mathbb{R}^n} \hat{u} \cdot v = \int_{\mathbb{R}^n} u \hat{v}$$

$$\circ \int_{\mathbb{R}^n} |u|^2 = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} |\hat{u}|^2 \quad \text{ID. PLANCHEREL}$$

$$\circ \widehat{u \times v} = \hat{u} \cdot \hat{v}$$

$$\circ \widehat{u \cdot v} = \frac{1}{(2\pi)^n} \hat{u} \times \hat{v}$$

Trasformate di Fourier in $L^2(\mathbb{R}^m)$

Come definire $\mathcal{F}$?

$$\hat{u}(\xi) = \int_{\mathbb{R}^m} \frac{u(x) e^{-i\xi x}}{dx}$$

molto posto se $u \in L^2(\mathbb{R}^m)$

$$|\hat{u}(\xi)| \leq \int_{\mathbb{R}^m} |u(x)| dx < +\infty$$

$\uparrow$
se $u \in L^1(\mathbb{R}^m)$

Hölder $\Rightarrow L^2(A) \subseteq L^1(A)$ se $m(A) < +\infty$.

Esempio $u(x) = \frac{\sin x}{x} \in L^2(\mathbb{R}) \setminus \underline{L^1(\mathbb{R})}$

Def. Sia $u \in L^2(\mathbb{R}^n)$, e sia $\{u_h\} \subseteq \mathcal{F}(\mathbb{R}^n)$
 tale che $u_h \rightarrow u$ in $L^2(\mathbb{R}^n)$.

Considero $\hat{u}_h \subseteq \mathcal{F}(\mathbb{R}^n)$, e definiamo

$$\overset{\circ}{u} := \lim_h \hat{u}_h \quad (\text{in } L^2(\mathbb{R}^n)).$$

IDEA:

$$\begin{array}{ccc} L^1 \supseteq \mathcal{F} \supseteq u_h & \xrightarrow{h \rightarrow +\infty} & u \in L^2(\mathbb{R}^n) \\ \mathcal{F} \downarrow \nabla & & \\ \mathcal{F} \supseteq \hat{u}_h & \xrightarrow{h \rightarrow +\infty} & \overset{\circ}{u} \\ & & u \end{array}$$

Observation:

1) $\exists \lim_n \hat{u}_n$ (in $L^2(\mathbb{R}^n)$)
 perché L^2 Banach e $\hat{u}_n$ è d. Cauchy

$$\|\hat{u}_h - \hat{u}_k\|_{L^2}^2 = \|\widehat{u_h - u_k}\|_{L^2}^2 \stackrel{\text{Plancherel}}{=} \frac{1}{(2\pi)^n} \|u_h - u_k\|_{L^2}^2 < \varepsilon$$

per $h, k \gg 1$, perché $\{u_h\}$ d. Cauchy

2) $\hat{u}$ non dipende dalla scelta di $\{u_h\}$:

$$u_h \xrightarrow{L^2} u, \quad v_h \xrightarrow{L^2} u \Rightarrow u_h - v_h \xrightarrow{L^2} u - u = 0 \Rightarrow \underbrace{\hat{u}_h - \hat{v}_h}_{\widehat{u_h - v_h}} \xrightarrow{L^2} 0$$

$\uparrow$
Plancherel

$\Rightarrow \hat{u}_h$ e $\hat{v}_h$ hanno lo stesso limite L^2 .

3) Se $u \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$, allora $\hat{u} = \hat{\hat{u}}$.

Prendiamo $\{u_h\} \subseteq \mathcal{S}(\mathbb{R}^n)$ tale che

$$\begin{cases} u_h \xrightarrow{L^2} u & \Rightarrow \hat{u}_h \xrightarrow{L^2} \hat{u} \quad (\text{per def. di } \hat{u}) \\ u_h \xrightarrow{L^1} u & \Rightarrow \hat{u}_h \xrightarrow{L^\infty} \hat{u} \quad (\text{perché } \mathcal{F}: L^1 \rightarrow L^\infty \text{ continuo}) \end{cases}$$

(esiste, tramite convoluzione con mollificatori)

$$\Rightarrow \hat{u} = \hat{u} \quad \text{q.o. su } \mathbb{R}^n.$$

4) L^1 id. di Plancherel:

$$\|u\|_{L^2}^2 = \frac{1}{(2\pi)^n} \|\hat{u}\|_{L^2}^2 \quad \forall u \in L^2(\mathbb{R}^n)$$

Suffici, prendendo $\{u_h\} \subseteq \mathcal{S}(\mathbb{R}^n) : u_h \xrightarrow{L^2} u.$

$$\|u_h\|_{L^2}^2 = \frac{1}{(2\pi)^n} \|\hat{u}_h\|_{L^2}^2 \quad \forall h$$

$$\begin{array}{ccc} \text{perché} & & \text{perché} \\ u_h \rightarrow u \text{ in } L^2 & \downarrow & \hat{u}_h \rightarrow \hat{u} \text{ in } L^2 \\ \|\hat{u}\|_{L^2}^2 & = & \frac{1}{(2\pi)^n} \|\hat{u}\|_{L^2}^2 \end{array}$$

5) Come scegliere $\hat{u} \in L^2(\mathbb{R})$. ($n=1$).

$$\text{Prendiamo } u_h = u \cdot \chi_{(-h, h)} = \begin{cases} u & \text{su } (-h, h) \\ 0 & \text{fuori} \end{cases}$$

Si ha $u_h \rightarrow u$ in $L^2(\mathbb{R})$

(per ev. puntuale + ev. dominata)

Per Plancherel (in L^2),

$$\|\hat{u}_h - \hat{u}\|_2 = \frac{1}{2\pi} \|u_h - u\|_2$$

$\downarrow$
0

$\downarrow$
0

$$\hat{u}_h \rightarrow \hat{u} \text{ in } L^2(\mathbb{R})$$

Quindi:

$$\hat{u}(\xi) = \lim_{h \rightarrow +\infty} \hat{u}_h(\xi) = \lim_{h \rightarrow +\infty} \int_{\mathbb{R}} u_h(x) e^{-i\xi x} dx$$

$$\boxed{\hat{u}(\xi) = \lim_{h \rightarrow +\infty} \int_{-h}^h u(x) e^{-i\xi x} dx}$$

$$u_h \in L^2(\mathbb{R}) \cap \underbrace{L^1(\mathbb{R})}$$

$$\uparrow \text{perché } \int_{\mathbb{R}} |u_h| = \int_{-h}^h |u_h| < +\infty$$

e vale pr. immersione

$$L^2([-h, h]) \subseteq L^1([-h, h]).$$

Esempio $\underline{u(x)} = \frac{\sin x}{x} \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$

$$\hat{u}(\xi) = \lim_{h \rightarrow +\infty} \int_{-h}^h \frac{\sin x}{x} e^{-i\xi x} dx = \pi \chi_{(-1, 1)}(\xi).$$

Proposizione Se $u \in L^2$, allora $\hat{u} \in L^2$ (per def!)

vale

$$\boxed{\check{u} = \frac{1}{(2\pi)^n} \hat{\hat{u}} \quad \forall x \in \mathbb{R}^n}$$

Dim.

Sia $\{u_h\} \subseteq \mathcal{S}(\mathbb{R}^n): u_h \xrightarrow{L^2} u$

Vale formula inversione in $\mathcal{S}$:

$$\check{u}_h = \frac{1}{(2\pi)^n} \hat{\hat{u}}_h \quad \forall h$$

Basta passare al lim per $h \rightarrow +\infty$

$$u_h \xrightarrow{L^2} u \quad \Rightarrow \quad \check{u}_h \xrightarrow{L^2} \check{u}$$

$$\check{u} = \frac{1}{(2\pi)^n} \hat{\hat{u}}$$

$$u_h \xrightarrow{L^2} u \quad \Rightarrow \quad \|u_h - u\|_2^2 \rightarrow 0$$

$$\Rightarrow_P \|\hat{u}_h - \hat{u}\|_2^2 \rightarrow 0$$

$$\Rightarrow_P \|\hat{\hat{u}}_h - \hat{\hat{u}}\|_2^2 \rightarrow 0$$

□

Osservazione Plancherel si può usare per
il calcolo di norme L^2

$$\int_{\mathbb{R}} \left| \frac{\sin x}{x} \right|^2 dx = \frac{1}{2\pi} \|\widehat{u}\|_{L^2}^2$$

$$= \frac{1}{2\pi} \int_{\mathbb{R}} \pi^2 \chi_{(-1,1)}^2(x) dx = \frac{1}{2\pi} \cdot \pi^2 \cdot 2 = \pi$$

Riassunto:

$$f: L^1 \longrightarrow L^\infty$$

$$f: \mathcal{S} \longrightarrow \mathcal{S}$$

$$f: L^2 \longrightarrow L^2$$

$$f: \mathcal{S}' \longrightarrow \mathcal{S}'$$

IDEA POSSIBILE : Data $T \in \mathcal{D}'(\mathbb{R}^n)$

$$\langle \hat{T}, \varphi \rangle \neq \langle T, \hat{\varphi} \rangle \quad \forall \varphi \in \mathcal{D}(\mathbb{R}^n)$$

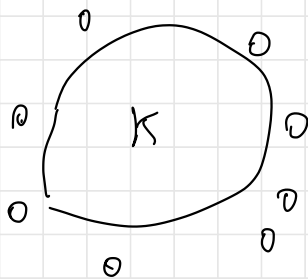
(suggerita dalla proprietà di f in $\mathcal{F}(\mathbb{R}^n)$)

$$\int_{\mathbb{R}^n} \hat{u} \varphi = \int_{\mathbb{R}^n} u \hat{\varphi} \quad \forall u, \varphi \in \mathcal{F}(\mathbb{R}^n)$$

PROBLEMA $\varphi \in \mathcal{D}(\mathbb{R}^n) \not\Rightarrow \hat{\varphi} \in \mathcal{D}(\mathbb{R}^n)$

In generale $\varphi \in \mathcal{D}(\mathbb{R}^n) \Rightarrow \hat{\varphi}$ analitica

e pertanto non può avere supporto compatto.



Recall:

Una successione $\{\varphi_h\} \subseteq \mathcal{D}(\mathbb{R}^n)$ converge a 0 in $\mathcal{D}(\mathbb{R}^n)$ se:

(1) $\exists K$ compatto : $\text{supp}(\varphi_h) \subseteq K \quad \forall h$

(2) $x^\alpha \varphi_h \rightarrow 0$ unif. su $K, \forall \alpha$

Def. Data $\{\varphi_h\} \subseteq \mathcal{S}(\mathbb{R}^n)$, $\varphi_h \rightarrow 0$ in $\mathcal{S}(\mathbb{R}^n)$

se:

$$x^\alpha D^\beta \varphi_h \rightarrow 0 \text{ unif. su } \mathbb{R}^n.$$

Dim. $\mathcal{D}(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n)$.

Per successioni in $\mathcal{D}(\mathbb{R}^n)$, che relazione c'è tra:

$$\varphi_h \rightarrow 0 \text{ in } \mathcal{D} \quad \boxed{\Rightarrow} \quad \varphi_h \rightarrow 0 \text{ in } \mathcal{S}$$

Sapete se $\varphi_h \rightarrow 0$ in $\mathcal{D}$, $\exists K$ compatto t.c.

$\varphi_h = 0$ fuori da K , e $D^\beta \varphi_h \rightarrow 0$ unif $\forall \beta$

Quindi $\forall \alpha, \beta$

$$\sup_{x \in \cancel{\mathbb{R}^n}} |x^\alpha D^\beta \varphi_h| \leq \sup_{x \in K} M_\alpha |D^\beta \varphi_h| \rightarrow 0$$

$\cancel{K}$

$$\varphi_h(x) = e^{-h} \cdot \varphi(x-h)$$

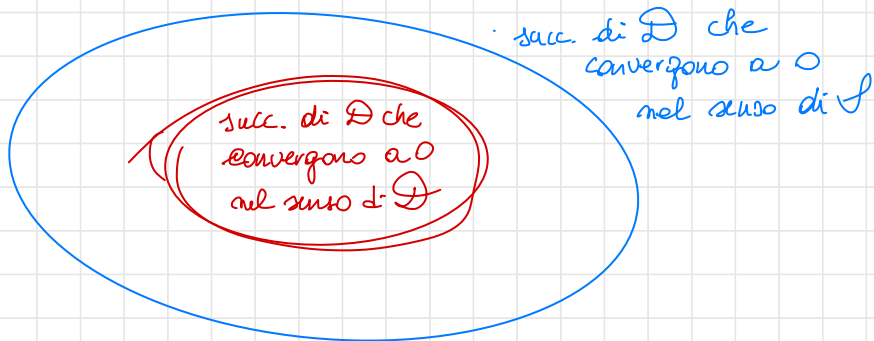
$$\varphi \in \mathcal{D}(\mathbb{R})$$

$$x^\alpha D^\beta \varphi_h = e^{-h} \cdot x^\alpha D^\beta \varphi(x-h)$$

$$\sup_{\mathbb{R}} x^\alpha D^\beta \varphi_h = e^{-h} \cdot \underbrace{\sup_{\mathbb{R}} x^\alpha D^\beta \varphi(x-h)}_{\substack{e^{-h} \cdot h^{\alpha/\alpha} \\ e^{-h} \cdot h^\alpha M_\beta}} \longrightarrow 0$$

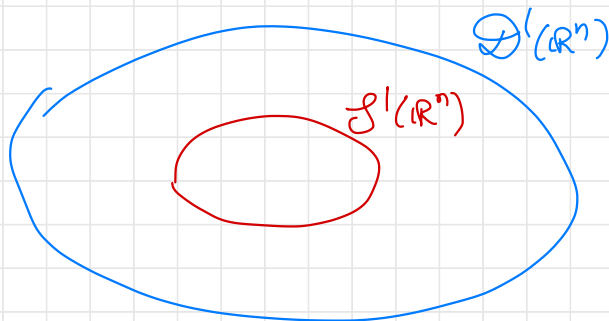
$$\text{Quindi } \varphi_h \rightarrow 0 \text{ in } \mathcal{S}(\mathbb{R})$$

$$\text{Ma } \varphi_h \not\rightarrow 0 \text{ in } \mathcal{D}(\mathbb{R})$$



Def. Una **DISTRIBUZIONE TEMPERATA** è una distribuzione $T \in \mathcal{D}'(\mathbb{R}^n)$ tale che
 $\{\varphi_h\} \subseteq \mathcal{D}(\mathbb{R}^n) : \varphi_h \rightarrow 0 \text{ in } \mathcal{S}(\mathbb{R}^n) \Rightarrow \langle T, \varphi_h \rangle \rightarrow 0.$

Def. $\mathcal{S}'(\mathbb{R}^n) = \{\text{distribuzioni temperate}\}$



sottospazio
vettoriale di $\mathcal{S}'(\mathbb{R}^n)$
(proprio)

$$T = T_\mu \quad \mu \in L^p(\mathbb{R}) \quad \text{e } \mu \text{ temperata}$$

$$\langle T_\mu, \varphi \rangle = \int_{\mathbb{R}} \mu \varphi \, dx$$